Email Spam Classification

Using NLP and Machine Learning

Support Vector Machine · TF-IDF · SpamAssassin Corpus

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Presentation Outline

01 **Problem Statement**
Why spam detection matters & project goals

02 **Dataset Overview**
SpamAssassin corpus — 4,198 emails, class distribution

03 **Methodology & Pipeline**
Preprocessing → TF-IDF → Model Training

04 **Key Results**
Naive Bayes vs SVM · Accuracy · Confusion matrix

05 **Live Application**
Interactive Streamlit web app screenshots

06 **Conclusions & Future Work**
Takeaways, limitations, and next steps

Problem Statement & Motivation

Phishing & Malware

Spam is a primary vector for phishing attacks, credential theft, and malware distribution.

Productivity Loss

Users lose significant time managing unsolicited email clutter every day.

False Positives Matter

Misclassifying legitimate emails as spam erodes user trust — precision is critical.

Project Goal

Build an accurate, interpretable spam filter using classical NLP & ML with high precision.

45%

of global email traffic is spam

Dataset Overview

Source: SpamAssassin Public Corpus · kaggle.com/datasets/beatoa/spamassassin-public-corpus

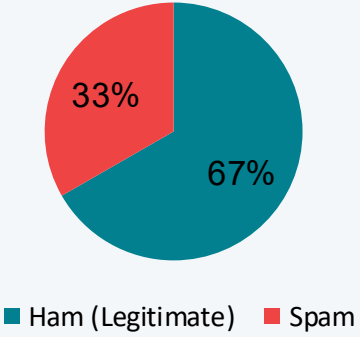
4,198
Total Emails

2,801
Ham (66.7%)

1,397
Spam (33.3%)

80/20
Train/Test Split

Feature	Configuration
Vocabulary Size	10,000 TF-IDF features
N-gram Range	Unigrams + Bigrams (1,2)
TF Scaling	Sublinear TF scaling enabled
Training set	3,358 emails
Test set	840 emails



Methodology & NLP Pipeline

1	2	3	4	5	6	7
Raw Email Input	Header Stripping	Text Normalisation	Stopword Removal	Lemma-tisation	TF-IDF Vectorisation	SVM Classifier
Headers + Body	Isolate body text	Lowercase · URLs → token	NLTK English list	WordNet Lemmatizer	10k features (1,2)-gram	LinearSVC + Platt scaling

Two Classifiers Compared

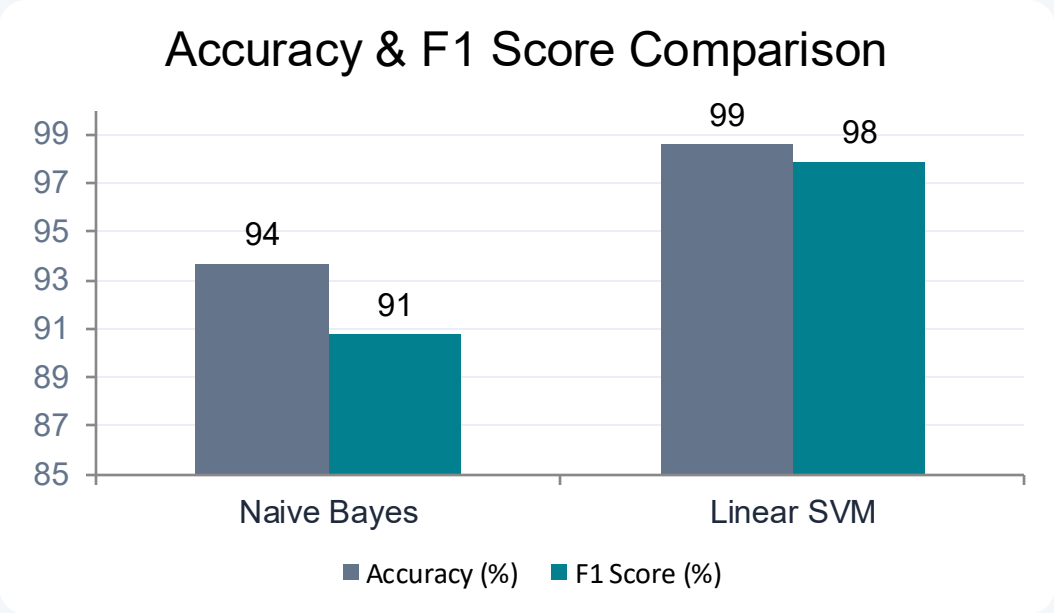
Multinomial Naive Bayes

Accuracy: 93.7% · F1: 90.8%
False Positives: 33 · False Negatives: 20

Linear SVM (LinearSVC)

Accuracy: 98.6% · F1: 97.9%
False Positives: 10 · False Negatives: 2

Key Results & Model Comparison



Confusion Matrix — SVM

Actual Ham →	Actual Spam →
550 ✅ True Ham	10 ❌ False Spam
2 ❌ False Ham	278 ✅ True Spam

98.6%
Test Accuracy

97.9%
F1 Score

1.8%
False Positive Rate

0.7%
False Negative Rate

Model Performance — Detailed Metrics

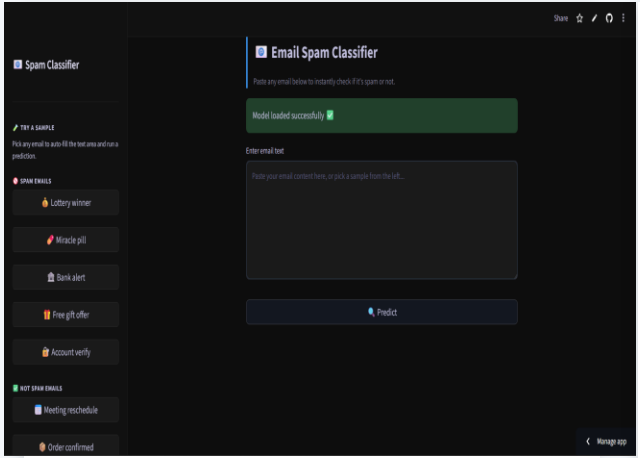
Metric	Naive Bayes	Linear SVM ⭐
Accuracy	93.7%	**98.6%**
F1 Score	90.8%	**97.9%**
False Positives	33	**10**
False Negatives	20	**2**
Probability Output	Native	Platt Scaling (CalibratedClassifierCV)

Why SVM Outperforms

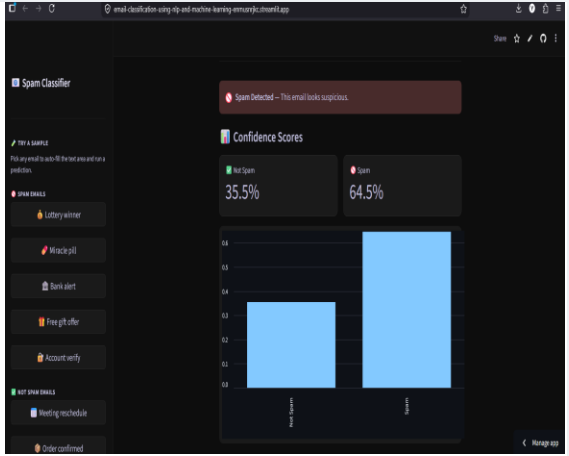
- SVM maximises the margin between classes, making it more robust to overlapping feature distributions.
- TF-IDF sparse high-dimensional spaces suit linear SVMs — they converge efficiently with LinearSVC.
- Naive Bayes assumes feature independence, which is violated by bigram features.
- SVM's 10 false positives vs NB's 33 — 70% reduction, crucial for user trust.
- Platt scaling provides calibrated probabilities for confidence scoring in the web app.

LinearSVC + TF-IDF (10k features, 1-2 grams) · Trained on 3,358 emails · Evaluated on 840 held-out emails

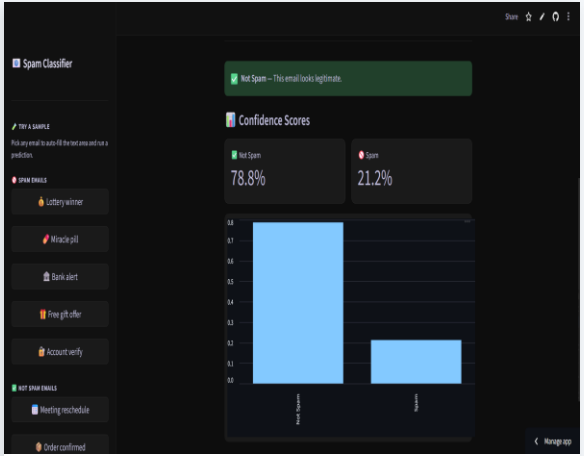
Deployed Streamlit Web Application



Home — Input Interface



Spam Detection Result



Ham Detection Result

Application Features

- Paste raw email text (including headers) for instant classification
- Visual confidence bar showing ham/spam probability score
- One-click sample emails for quick demo testing
- Expandable preprocessed text view for model transparency

🔗 Live Demo: https://email-classification-using-nlp-and-machine-learning-enmusnrjkc.streamlit.app/

Conclusions & Limitations

✅ Key Conclusions

1. Linear SVM + TF-IDF achieves 98.6% accuracy on the SpamAssassin corpus
2. 70% fewer false positives vs Naive Bayes — critical for user trust
3. Classical NLP pipeline is highly effective for email spam detection
4. Platt scaling enables calibrated confidence scores for UI display
5. End-to-end deployment demonstrates production readiness of the model

⚠️ Limitations

- Static model — requires periodic retraining as spam evolves
- English-only — no multilingual or Unicode spam handling
- No adversarial robustness — obfuscated spam may bypass detection
- Dataset skew: 66.7% ham may bias towards reducing false positives
- No image or attachment-based spam detection (text-only pipeline)

Future Work & Enhancements

Transformer-Based Models

Fine-tune BERT or DistilBERT for context-aware spam detection and improved accuracy on subtle phishing emails.

Multilingual Support

Extend the pipeline to handle non-English emails using multilingual embeddings and language detection.

Online / Active Learning

Implement incremental learning so the model adapts continuously as new spam patterns emerge in real-time.

Multi-Modal Detection

Analyse embedded images, attachments, and HTML structure for richer feature extraction beyond plain text.

Adversarial Robustness

Test against obfuscated spam (character substitution, homoglyphs) and train with adversarial augmentation.

Explainability (LIME/SHAP)

Integrate LIME or SHAP to generate per-email feature importance explanations in the web app.

Thank You

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github.com · Live Demo: streamlit.app